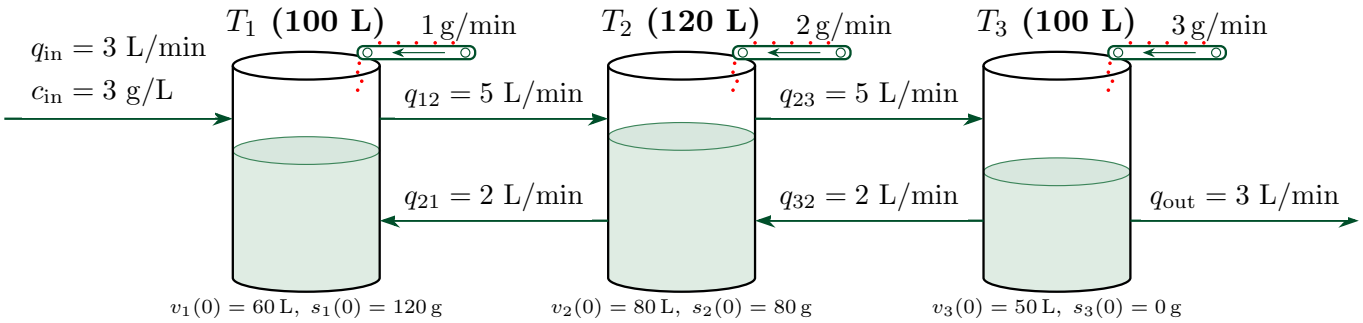
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MA3521 — In-Class Tank Worksheet Key

Name:

1.



Tanks are well-mixed, t in min, $\mathbf{v}(t)$ in L, $\mathbf{s}(t)$ in g, and conveyers add dry salt at the rate shown.

(a) $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \begin{pmatrix} 120 \\ 80 \\ 0 \end{pmatrix}$ g $\quad 0 < t <$ $\boxed{\infty}$ min

$\mathbf{v}(t) = \boxed{\begin{pmatrix} 60 \\ 80 \\ 50 \end{pmatrix}}$ L $\qquad \mathbf{b} = \boxed{\begin{pmatrix} 10 \\ 2 \\ 3 \end{pmatrix}}$ g/min

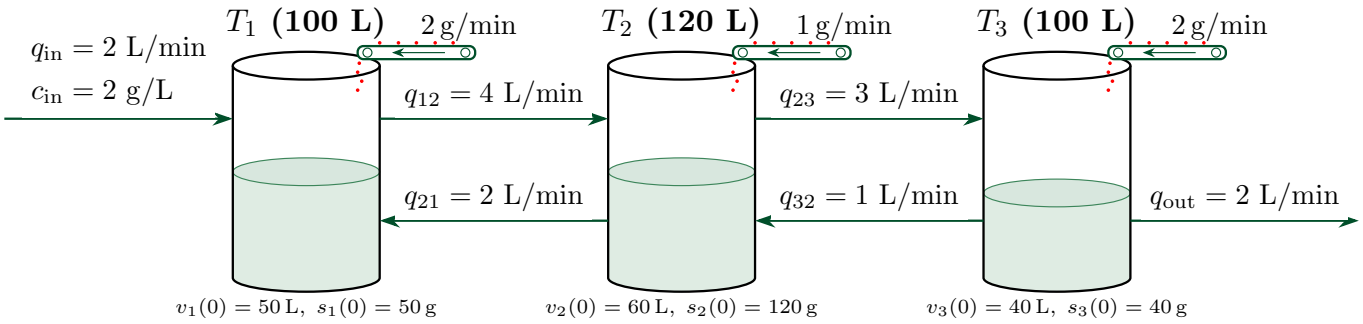
(b) $\boldsymbol{A} = \boxed{\begin{pmatrix} -5/60 & 2/80 & 0 \\ 5/60 & -7/80 & 2/50 \\ 0 & 5/80 & -5/50 \end{pmatrix}}$ min^{-1}

(c) A UC form is $\mathbf{s}_p(t) = \boxed{\mathbf{a}}$. $\qquad \mathbf{0} = \boldsymbol{A}\mathbf{a} + \mathbf{b} \implies \boldsymbol{A}\mathbf{a} = -\mathbf{b}$

Constants are determined by row reducing $\boldsymbol{M}_{\text{UC}} = \boxed{\left[\begin{array}{ccc|c} -5/60 & 2/80 & 0 & -10 \\ 5/60 & -7/80 & 2/50 & -2 \\ 0 & 5/80 & -5/50 & -3 \end{array}\right]}$

Volume check: $\quad 3 + 2 = 5, \quad 5 + 2 = 2 + 5, \quad 5 = 2 + 3 \quad$ (L/min)

2.



Tanks are well-mixed, t in min, $\mathbf{v}(t)$ in L, $\mathbf{s}(t)$ in g, and conveyers add dry salt at the rate shown.

(a) $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \begin{pmatrix} 50 \\ 120 \\ 40 \end{pmatrix}$ g $\quad 0 < t <$ $\boxed{\infty}$ min

$\mathbf{v}(t) =$ $\boxed{\begin{pmatrix} 50 \\ 60 \\ 40 \end{pmatrix}}$ L $\qquad \mathbf{b} =$ $\boxed{\begin{pmatrix} 6 \\ 1 \\ 2 \end{pmatrix}}$ g/min

(b) $\boldsymbol{A} =$ $\boxed{\begin{pmatrix} -4/50 & 2/60 & 0 \\ 4/50 & -5/60 & 1/40 \\ 0 & 3/60 & -3/40 \end{pmatrix}}$ min^{-1}

(c) A UC form is $\mathbf{s}_p(t) =$ $\boxed{\mathbf{a}}$. $\qquad \mathbf{0} = \boldsymbol{A}\mathbf{a} + \mathbf{b} \implies \boldsymbol{A}\mathbf{a} = -\mathbf{b}$

Constants are determined by row reducing $\boldsymbol{M}_{\text{UC}} =$ $\boxed{\left[\begin{array}{ccc|c} -4/50 & 2/60 & 0 & -6 \\ 4/50 & -5/60 & 1/40 & -1 \\ 0 & 3/60 & -3/40 & -2 \end{array}\right]}$

(d) Not on exam. Solve the IVP $\mathbf{y} =$ $\boxed{\begin{array}{c} (1025, 1380, 1080)^T/6 \\ + \exp(\boldsymbol{A}t)(-725, -660, -840)^T/6 \end{array}}$

Volume check $2 + 2 = 4$, $\quad 4 + 1 = 2 + 3$, $\quad 3 = 1 + 2$ $\quad$ (L/min)